

Project on Hamiltonian Simulation - Part II

Theory Exercises

Exercise 1: An Algorithm for Trotterized Hamiltonian Simulation

Exercise 1.1: Algorithm Sketch

Recall the Trotter formula:

$$e^{-iHt} \approx \left(\prod_P e^{-ia_P Pt/r} \right)^r \quad (1)$$

Given the helper function `qc.pauli(P, theta)` implementing $e^{iP\theta}$, consider the following pseudocode:

```
procedure trotter_approx(H, r, t):
    # H is a list of (coefficient, pauli_string) pairs;
    # initialise quantum circuit with n qubits,
    # with n = Pauli string length
    n = len(H[0][1])
    qc = quantum_circuit(n)

    for step in range(r):
        for (a_P, P) in H:
            theta = -a_P * t / r
            qc.pauli(P, theta)

    return qc
```

Explanations:

- The outer range(r) ensures the power of r is applied
- The product over P, or application of all gates, is done in the inner loop using the helper function
- For theta, we need $e^{-ia_P Pt/r}$, so we set $\theta = -a_P t/r$ so that $e^{iP\theta} = e^{iP(-a_P t/r)} = e^{-ia_P Pt/r}$

Exercise 1.2: Why Pauli Decomposition Format?

There is a dimensionality problem when providing the full n qubit Hamiltonian.

Since the Hamiltonian is a matrix of dimensions $2^n \times 2^n$, the number of entries to store is $O(4^n)$. For large n (e.g. $n > 30$), this gets exponentially large and becomes impossible to store!

On the other hand, for a Pauli decomposition of length n , and a Hamiltonian H of $\text{poly}(n)$ terms, we store $O(2 * \text{poly}(n) * n) = O(\text{poly}(n))$ terms. This is much more manageable, as it is polynomial in length (and not exponential)!

The circuit then has $m e^{-ia_P P t/r}$ exponentials constructed with $O(n)$ gates, repeated r times, yielding complexity $O(mrn)$, polynomial in size. This is much better than the exponential complexity of the Hamiltonian derived earlier.

Exercise 2: The Transverse-Field Ising Hamiltonian

Recall the transverse-field Ising Hamiltonian formula:

$$H_{\text{Ising}} = -J \sum_{i=0}^{n-1} Z_i Z_{i+1 \pmod n} - h \sum_i X_i \quad (2)$$

2.1 Interaction Terms

Part (a): proof that $Z_i Z_j$ is diagonal

Lemma: The tensor product of diagonal matrices is diagonal.

Proof: Let $A = \text{diag}(a_1, \dots, a_m)$ and $B = \text{diag}(b_1, \dots, b_n)$ be diagonal matrices.

We write the Kronecker product $A \otimes B$ in block form:

$$A \otimes B = \begin{pmatrix} A_{11}B & A_{12}B & \cdots & A_{1m}B \\ A_{21}B & A_{22}B & \cdots & A_{2m}B \\ \vdots & \vdots & \ddots & \vdots \\ A_{m1}B & A_{m2}B & \cdots & A_{mm}B \end{pmatrix} \quad (3)$$

Since A is diagonal, $A_{ij} = 0$ for $i \neq j$, so all off-diagonal blocks vanish:

$$A \otimes B = \begin{pmatrix} a_1 B & 0 & \cdots & 0 \\ 0 & a_2 B & \cdots & 0 \\ \vdots & \vdots & \ddots & \vdots \\ 0 & 0 & \cdots & a_m B \end{pmatrix} \quad (4)$$

Since B is also diagonal, each block $a_i B$ is diagonal. Therefore, $A \otimes B$ is block-diagonal with diagonal blocks, making the entire matrix diagonal. $\square$

Proof of part (a): By the lemma, since I and Z are diagonal, and Z_i, Z_j are tensor products of diagonal matrices, they are both diagonal in the computational basis.

Part (b): Action on computational basis states

We compute how $e^{i(Jt/r)Z_i Z_{i+1}}$ acts on basis states $|a\rangle_i |b\rangle_{i+1}$ for $a, b \in \{0, 1\}$.

First, recall that the Pauli Z matrix acts as:

$$Z |0\rangle = |0\rangle \quad (5)$$

$$Z |1\rangle = -|1\rangle \quad (6)$$

For the product $Z_i Z_{i+1}$:

$$(Z_i Z_{i+1}) |0\rangle_i |0\rangle_{i+1} = Z_i |0\rangle \otimes Z_{i+1} |0\rangle = |0\rangle \otimes |0\rangle = (+1) |00\rangle \quad (7)$$

$$(Z_i Z_{i+1}) |0\rangle_i |1\rangle_{i+1} = Z_i |0\rangle \otimes Z_{i+1} |1\rangle = |0\rangle \otimes (-|1\rangle) = (-1) |01\rangle \quad (8)$$

$$(Z_i Z_{i+1}) |1\rangle_i |0\rangle_{i+1} = Z_i |1\rangle \otimes Z_{i+1} |0\rangle = (-|1\rangle) \otimes |0\rangle = (-1) |10\rangle \quad (9)$$

$$(Z_i Z_{i+1}) |1\rangle_i |1\rangle_{i+1} = Z_i |1\rangle \otimes Z_{i+1} |1\rangle = (-|1\rangle) \otimes (-|1\rangle) = (+1) |11\rangle \quad (10)$$

The eigenvalues are $\lambda_{ab} = (-1)^{a+b}$: $+1$ when $a = b$, and -1 when $a \neq b$.

From Part I Exercise 1.1, we proved that for a diagonal matrix $D = \text{diag}(\lambda_1, \dots, \lambda_n)$:

$$e^D = \text{diag}(e^{\lambda_1}, \dots, e^{\lambda_n}) \quad (11)$$

Since $Z_i Z_{i+1}$ is diagonal with entries equal to the eigenvalues computed above, we apply the exponential:

$$e^{i(Jt/r)Z_i Z_{i+1}} |a\rangle_i |b\rangle_{i+1} = e^{i(Jt/r)\lambda_{ab}} |a\rangle_i |b\rangle_{i+1} = e^{i(Jt/r)(-1)^{a+b}} |a\rangle_i |b\rangle_{i+1} \quad (12)$$

Expanding, we get:

$$e^{i(Jt/r)Z_i Z_{i+1}} |0\rangle |0\rangle = e^{i(Jt/r)} |0\rangle |0\rangle \quad (13)$$

$$e^{i(Jt/r)Z_i Z_{i+1}} |0\rangle |1\rangle = e^{-i(Jt/r)} |0\rangle |1\rangle \quad (14)$$

$$e^{i(Jt/r)Z_i Z_{i+1}} |1\rangle |0\rangle = e^{-i(Jt/r)} |1\rangle |0\rangle \quad (15)$$

$$e^{i(Jt/r)Z_i Z_{i+1}} |1\rangle |1\rangle = e^{i(Jt/r)} |1\rangle |1\rangle \quad (16)$$

2.2 Transverse Field Terms

Part (a): Diagonalizing basis

Recall the Pauli X matrix has eigenvectors:

$$|+\rangle = \frac{1}{\sqrt{2}}(|0\rangle + |1\rangle), \quad \text{eigenvalue } +1 \quad (17)$$

$$|-\rangle = \frac{1}{\sqrt{2}}(|0\rangle - |1\rangle), \quad \text{eigenvalue } -1 \quad (18)$$

Applying the transverse field term $e^{i(ht/r)X_i}$ to each state:

$$e^{i(ht/r)X_i} |+\rangle = \sum_{k=0}^{\infty} \frac{(i(ht/r)X_i)^k}{k!} |+\rangle \quad (19)$$

$$= \sum_{k=0}^{\infty} \frac{(i(ht/r))^k X_i^k}{k!} |+\rangle \quad (20)$$

$$= \sum_{k=0}^{\infty} \frac{(i(ht/r))^k \cdot 1^k}{k!} |+\rangle \quad (\text{since } X_i |+\rangle = |+\rangle) \quad (21)$$

$$= \left(\sum_{k=0}^{\infty} \frac{(i(ht/r))^k}{k!} \right) |+\rangle \quad (22)$$

$$= e^{i(ht/r)} |+\rangle \quad (23)$$

Similarly:

$$e^{i(ht/r)X_i} |-\rangle = \sum_{k=0}^{\infty} \frac{(i(ht/r))^k X_i^k}{k!} |-\rangle \quad (24)$$

$$= \sum_{k=0}^{\infty} \frac{(i(ht/r))^k \cdot (-1)^k}{k!} |-\rangle \quad (\text{since } X_i |-\rangle = -|-\rangle) \quad (25)$$

$$= e^{-i(ht/r)} |-\rangle \quad (26)$$

Therefore $e^{i(ht/r)X_i}$ is diagonal in the X -basis $\{|+\rangle, |-\rangle\}$, with eigenvalues $e^{\pm i(ht/r)}$.

Part (b): Relation to rotation gate and bit flip

Recall the rotation gate's definition:

$$R_X(\theta) = e^{-i\theta X/2} \quad (27)$$

Compare with $e^{i(ht/r)X_i}$:

$$e^{i(ht/r)X_i} = e^{-i\theta X_i/2} \quad (28)$$

Matching exponents:

$$i(ht/r) = -i\theta/2 \quad \Rightarrow \quad \theta = -\frac{2ht}{r} \quad (29)$$

Therefore:

$$e^{i(ht/r)X_i} = R_{X_i}\left(-\frac{2ht}{r}\right) \quad (30)$$

When the rotation gate acts as a bit flip

The bit flip gate X is equivalent to a rotation of (any odd multiple of) π :

$$X = R_X(\pi) \quad (31)$$

For $e^{i(ht/r)X_i}$ to act as a bit flip:

$$-\frac{2ht}{r} = \pi + 2\pi k, \quad k \in \mathbb{Z} \quad (32)$$

Solving for $t \geq 0$

$$t = -\frac{\pi r(1 + 2k)}{2h} \quad (33)$$

For positive t , taking $k = -1, -2, -3, \dots$:

$$t = \frac{\pi r(2m - 1)}{2h}, \quad m = 1, 2, 3, \dots \quad (34)$$

Therefore, the transverse field term acts as a bit flip when:

$$t \in \left\{ \frac{\pi r}{2h}, \frac{3\pi r}{2h}, \frac{5\pi r}{2h}, \dots \right\} \quad (35)$$